Homework 10

3.2.2 Prove that for $\varphi \in \mathscr{S}(\mathbb{R}^n)$ and $1 \leqslant j,k \leqslant n$ and all $x \in \mathbb{R}^n$ we have

$$\partial_j \partial_k \varphi(x) = -R_j R_k \Delta\varphi(x).$$

Here $\Delta\varphi = \sum_{j=1}^n \partial_j^2 \varphi$ is the *Laplacian* of φ, given by multiplication by $-4\pi^2|\xi|^2$ on the Fourier transform. Conclude that

$$\|\partial_{j_1}...\partial_{j_{2m}}\varphi\|_{L^2(\mathbb{R}^n)} \leqslant \|\Delta^m \varphi\|_{L^2(\mathbb{R}^n)}$$

for all $m \in \mathbb{N}, j_1, ..., j_{2m} \in \{0, ..., n\}$ and all $\varphi \in \mathscr{S}(\mathbb{R}^n)$

Proof. Recall that the jth Riesz transform is given by $R_j(\varphi)(x) = \frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}(\varphi * W_{\Omega_j})(x)$, and that for every $\xi \in \mathbb{R}^n \setminus \{0\}$ we have

$$\frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}\widehat{(\varphi * W_{\Omega_j})}(\xi) = \frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}\widehat{W_{\Omega_j}}\widehat{\varphi}(\xi) = -i\frac{\xi_j}{|\xi|}\widehat{\varphi}(\xi).$$

This gives us

$$\begin{aligned}
\partial_j \partial_k \varphi(x) &= [\widehat{\partial_j \partial_n \varphi}]^\vee(x) \\
&= [(2\pi i \xi_j)(2\pi i \xi_k)\widehat{\varphi}(\xi)]^\vee(x) \\
&= [-4\pi^2 \xi_j \xi_k \widehat{\varphi}(\xi)]^\vee(x) \\
&= [-4\pi^2 \xi_j \xi_k \frac{\widehat{\Delta\varphi}}{-4\pi^2|\xi|^2}(\xi)]^\vee(x) \\
&= [i\frac{\xi_j}{|\xi|} \cdot \frac{-i\xi_k}{|\xi|} \cdot \widehat{\Delta\varphi}(\xi)]^\vee(x) = -R_j R_k \Delta\varphi(x).
\end{aligned}$$

And this holds for all x because $\varphi \in \mathscr{S}$, and hence $\widehat{\varphi}^\vee(x) = \varphi(x)$ everywhere. We must now show that

$$\|\partial_{j_1}...\partial_{j_{2m}}\varphi\|_{L^2(\mathbb{R}^n)} \leqslant \|\Delta^m \varphi\|_{L^2(\mathbb{R}^n)}.$$

By Plancherel, showing this holds for the Fourier transforms of these functions is equivalent. Notice that

$$(\partial_{j_1}...\partial_{j_{2m}}\varphi)\widehat{\ }(\xi) = (2\pi i)^{2m}\xi_{j_1}...\xi_{j_{2m}}\widehat{\varphi}(\xi). \tag{1}$$

$$(\Delta^m \varphi)\widehat{\ } = (-4\pi^2|\xi|^2)^m \widehat{\varphi}(\xi) = (-1)^m(2\pi|\xi|)^{2m}\widehat{\varphi}(\xi). \tag{2}$$

Taking absolute values, we get

$$|(1)| = (2\pi)^{2m}|\xi_{j_1}...\xi_{j_{2m}}||\widehat{\varphi}(\xi)|.$$
$$|(2)| = (2\pi)^{2m}|\xi|^{2m}|\widehat{\varphi}(\xi)|.$$

Hence, we must show that $|\xi_{j_1}...\xi_{j_{2m}}| \leqslant |\xi|^{2m}$ and we're done. But $|\xi|^{2m} = (\xi_1^2 + \xi_2^2 + ... + \xi_n^2)^m \geqslant |\xi_l|^{2m}$ for all $l \leqslant n$. $|\xi_{j_1}...\xi_{j_{2m}}|$ is largest whenever each $\xi_{j_k}, k \leqslant 2m$ is equal to $\max_{1\leqslant l\leqslant n}\{\xi_{j_l}\} = \xi_\mu$. In this case, $|\xi_{j_1}...\xi_{j_{2m}}| = |\xi_\mu|^{2m}$. But we already showed that $|\xi|^{2m}| \geqslant |\xi_l|^{2m}$ for $l \leqslant n$. This shows that $|\xi_{j_1}...\xi_{j_{2m}}| \leqslant |\xi|^{2m}$. □